

Digital Forensics Investigation Using Autopsy: Case Creation and Evidence Import

EXPERIMENT NO: 05

DATE:

1. AIM

To use the Autopsy digital forensics platform to create a new forensic case, import digital evidence (disk image files), configure ingest modules, and perform basic forensic analysis and reporting.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows, Linux, or macOS
- **Forensics Tool:** Autopsy Digital Forensics Platform
- **Evidence Files:** Disk image files (4Dell Latitude CPi.E01, 4Dell Latitude CPi.E02) downloaded from the provided forensic repository.

3. DESCRIPTION

Autopsy is a premier open-source digital forensics platform and graphical interface to The Sleuth Kit and other digital forensics tools. It is widely used by law enforcement, military, and corporate examiners to analyze and extract critical data from digital devices.

In a standard forensic investigation, raw data or disk images (such as .E01, .dd, or .raw formats) are imported into a securely managed "Case". Autopsy utilizes various **Ingest Modules** (like File Type Identification, Keyword Searching, Timeline Analysis, and Hash Lookups) to process this evidence in the background, automatically categorizing artifacts such as web history, file system metadata, and communication records for the examiner to review. Autopsy is the premier end-to-end open source digital forensics platform. Built by Sleuth Kit Labs with the core features you expect in commercial forensic tools, Autopsy is a fast, thorough, and efficient hard drive investigation solution that evolves with your needs.

4. PROCEDURE

Step 1: Installation & Preparation

1. Download and install the Autopsy application based on the host operating system.
2. Download the required evidence files (4Dell Latitude CPi.E01) to a secure local directory.
3. Launch the Autopsy application to view the welcome screen.

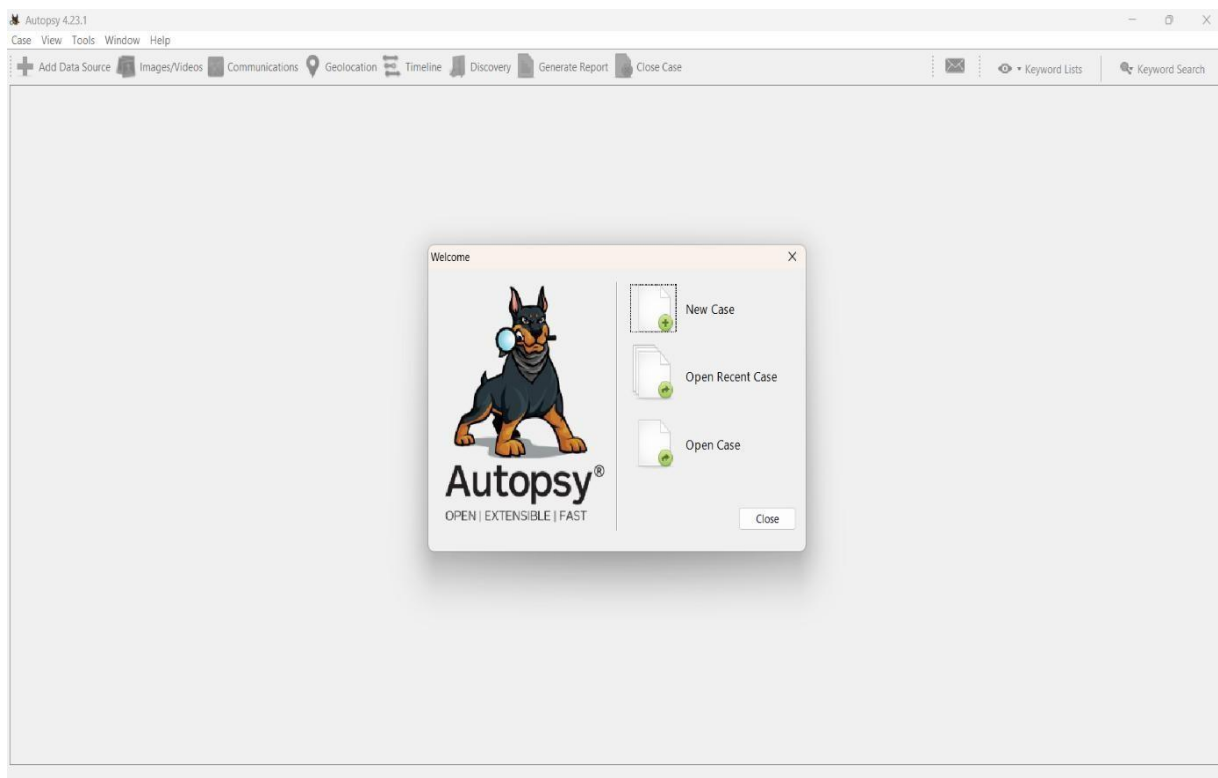
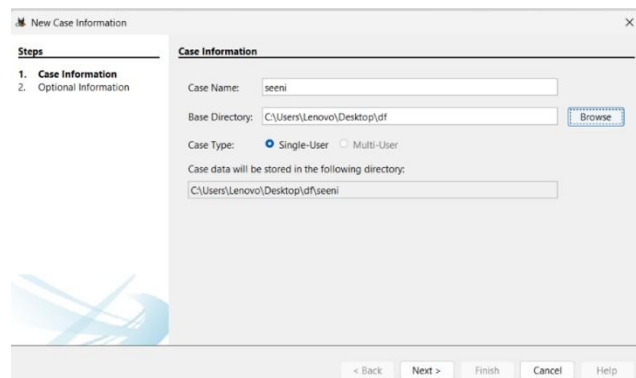


Figure 1: Autopsy startup menu and tool interface

Step 2: Starting a New Case

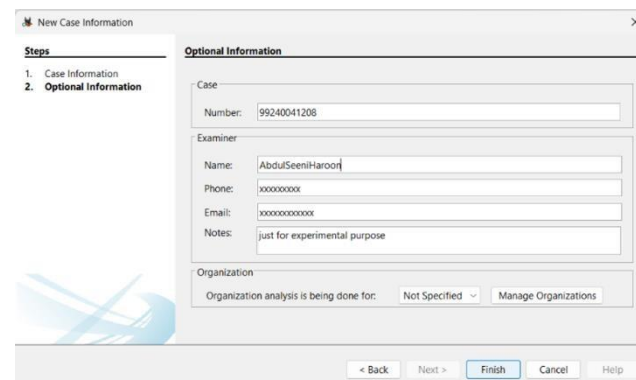
1. Click on **New Case** from the startup menu.
2. Enter the case name (e.g., EX05) and specify the base directory where the case data will be stored.
3. Click **Next** to proceed to the optional information.

4. Fill in the required case details, including the case number and examiner's name (seeni), then click **Finish**.



The screenshot shows the 'New Case Information' dialog box with the 'Case Information' tab selected. The 'Steps' pane on the left shows '1. Case Information' and '2. Optional Information'. The 'Case Information' section contains the following fields: 'Case Name' with the value 'seeni', 'Base Directory' with the value 'C:\Users\Lenovo\Desktop\df' and a 'Browse' button, 'Case Type' with 'Single-User' selected (radio button) and 'Multi-User' unselected, and 'Case data will be stored in the following directory:' with the value 'C:\Users\Lenovo\Desktop\df\seeni'. At the bottom are buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

Figure 2: Creating a new forensic case and setting the base directory



The screenshot shows the 'New Case Information' dialog box with the 'Optional Information' tab selected. The 'Steps' pane on the left shows '1. Case Information' and '2. Optional Information'. The 'Optional Information' section contains the following fields: 'Case Number' with the value '99240041208', 'Examiner Name' with the value 'AbdulSeeniHaroori', 'Phone' with the value 'xxxxxxxx', 'Email' with the value 'xxxxxxxxxxxx', 'Notes' with the value 'just for experimental purpose', and 'Organization' with a dropdown menu set to 'Not Specified' and a 'Manage Organizations' button. At the bottom are buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

Figure 3: Entering Optional Information including Examiner details

Step 3: Adding a Data Source

1. When prompted to add a data source, choose the type of data source (e.g., Disk Image or VM file).
2. Browse to the local directory and select the downloaded evidence file (4Dell Latitude CPi.E01).
3. **Configure Ingest Modules:** Select the specific analysis modules required for the investigation.
4. Click Next to start the analysis.

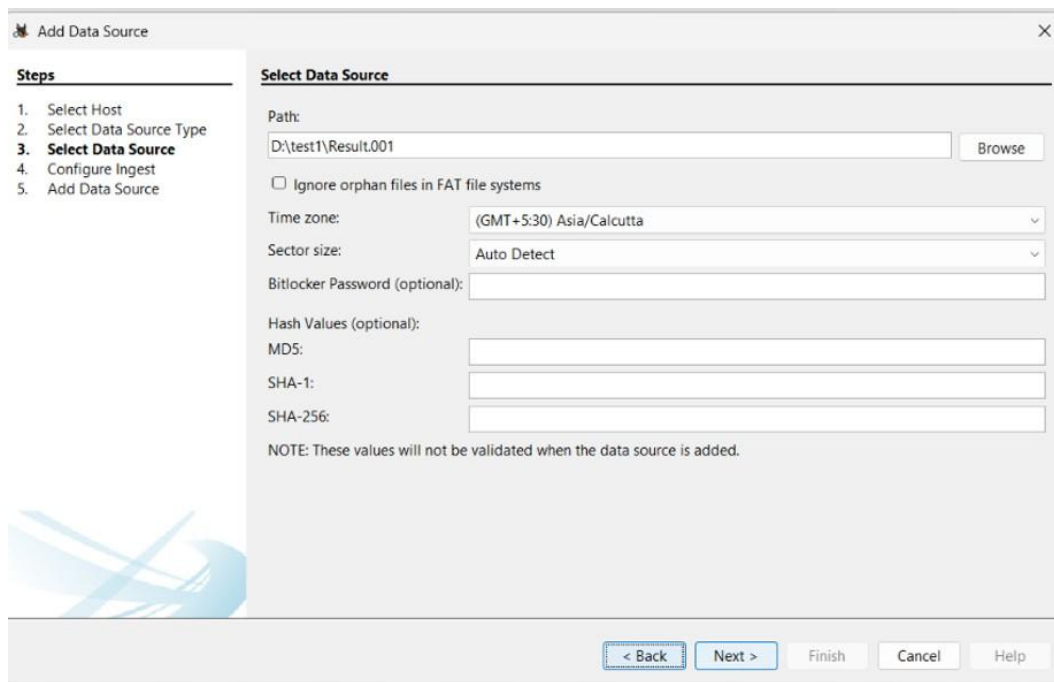


Figure 4: Importing the .E01 evidence file data source

Step 4: Initial Analysis and Overview

1. Monitor the **Ingest Progress** in the lower-left corner of the interface as Autopsy processes the data source.
2. Once processing yields results, use the **Tree Viewer** on the left pane to explore different categorized aspects like File System, Deleted Files, Data Artifacts, etc.

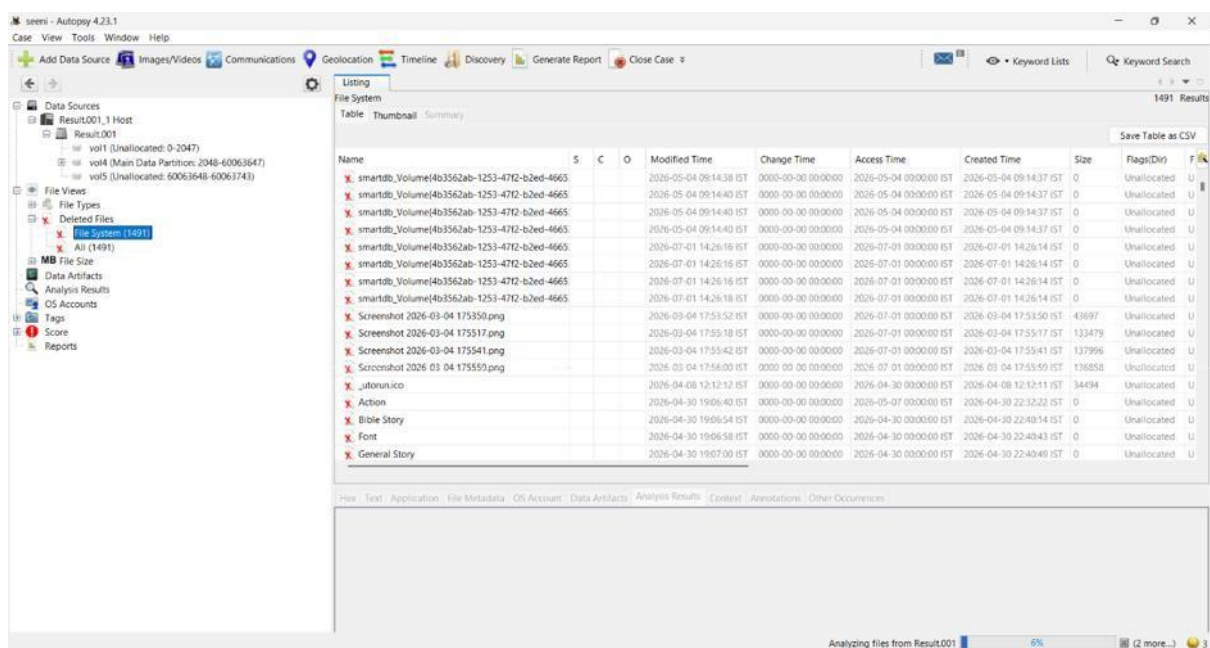


Figure 5: Exploring categorized forensic artifacts and deleted files in the Tree Viewer

Step 5: Detailed Analysis

1. **File Analysis:** Navigate through files and folders under the File Types or File System section. Open, view, or extract files for further examination.
2. Select an individual file (e.g., audio files under the Audio category) to view its hex, text, application, or file metadata in the lower pane.

The screenshot displays a software interface for file analysis. At the top, there's a 'Listing' tab and a sub-tab 'Audio' with 'Table', 'Thumbnail', and 'Summary' views. A 'Save Table as CSV' button is on the right. Below is a table with columns: Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, Flags(Dir), Flags(Meta), and K. The table lists various audio files like newalert.wav, newemail.wav, online.wav, alert.wav, analize.wav, pop.wav, ready.wav, stop.wav, and SOUND18.WAV. Below the table is a playback interface with tabs for Hex, Text, Application, File Metadata, OS Account, Data Artifacts, Analysis Results, Context, Annotations, and Other Occurrences. The 'Analysis Results' tab is active, showing a progress bar at 20% and a volume control slider.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	K
newalert.wav				2001-08-23 23:30:00 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	9306	Allocated	Allocated	ur
newemail.wav				2001-08-23 23:30:00 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	18052	Allocated	Allocated	ur
online.wav				2001-08-23 23:30:00 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	9306	Allocated	Allocated	ur
alert.wav				2002-06-15 01:48:30 IST	2004-08-25 21:26:06 IST	2004-02-18 15:30:03 IST	2004-02-16 16:21:17 IST	11370	Allocated	Allocated	ur
analize.wav				2002-06-15 01:46:22 IST	2004-08-25 21:26:06 IST	2004-02-18 15:15:36 IST	2004-02-16 16:21:17 IST	75812	Allocated	Allocated	ur
pop.wav				2002-06-15 01:44:28 IST	2004-08-25 21:26:07 IST	2004-02-18 15:11:03 IST	2004-02-16 16:21:17 IST	4486	Allocated	Allocated	ur
ready.wav				2002-06-18 00:09:08 IST	2004-08-25 21:26:07 IST	2004-02-18 15:16:07 IST	2004-02-16 16:21:17 IST	15920	Allocated	Allocated	ur
stop.wav				2002-06-15 01:43:42 IST	2004-08-25 21:26:06 IST	2004-02-18 14:53:47 IST	2004-02-16 16:21:17 IST	19040	Allocated	Allocated	ur
newalert.wav				2001-08-23 23:30:00 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	9306	Allocated	Allocated	ur
newemail.wav				2001-08-23 23:30:00 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	18052	Allocated	Allocated	ur
online.wav				2001-08-23 23:30:00 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	9306	Allocated	Allocated	ur
type.wav				2000-12-05 23:40:32 IST	2004-08-20 03:55:25 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	4454	Allocated	Allocated	ur
SOUND18.WAV				2001-08-23 23:30:00 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	3986	Allocated	Allocated	ur
SOUND38.WAV				2001-08-23 23:30:00 IST	2004-08-20 03:54:59 IST	2004-08-20 03:54:59 IST	2004-08-20 03:54:59 IST	13024	Allocated	Allocated	ur
PINBALL.MID				2001-08-23 23:30:00 IST	2004-08-20 03:54:54 IST	2004-08-20 03:54:54 IST	2004-08-20 03:54:54 IST	108607	Allocated	Allocated	ur
PINBALL.MID				2001-08-23 23:30:00 IST	2004-08-20 03:54:54 IST	2004-08-20 03:54:54 IST	2004-08-20 03:54:54 IST	28888	Allocated	Allocated	ur

Figure 6: Detailed analysis and viewing extracted files (e.g., Audio playback and metadata)

Step 6: Reporting and Case Closure

1. Click on **Generate Report** from the main toolbar.
2. Choose the desired report format (e.g., Excel Report, HTML Report) under the Select and Configure Report Modules window.
3. Click **Next** and select the specific result types to include (e.g., Accounts, E-Mail Messages, etc.).
4. Review the generated report to ensure all relevant information and exported artifacts are accurately documented.

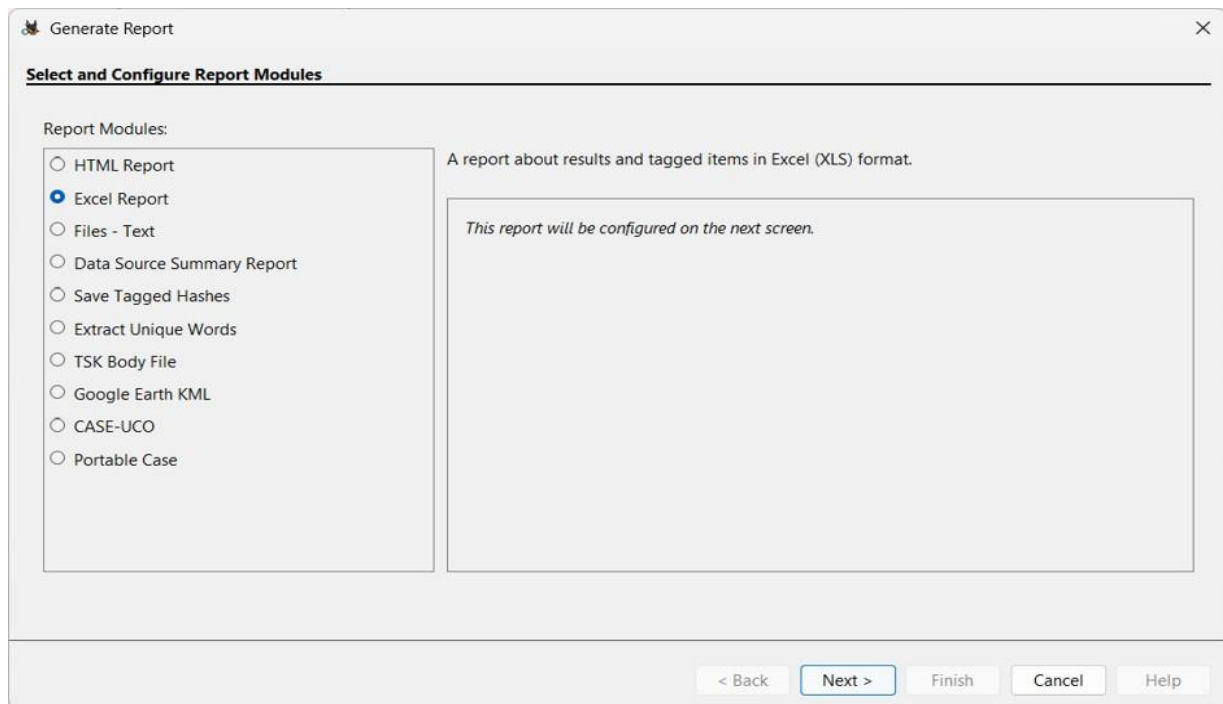


Figure 7: Selecting the report module format

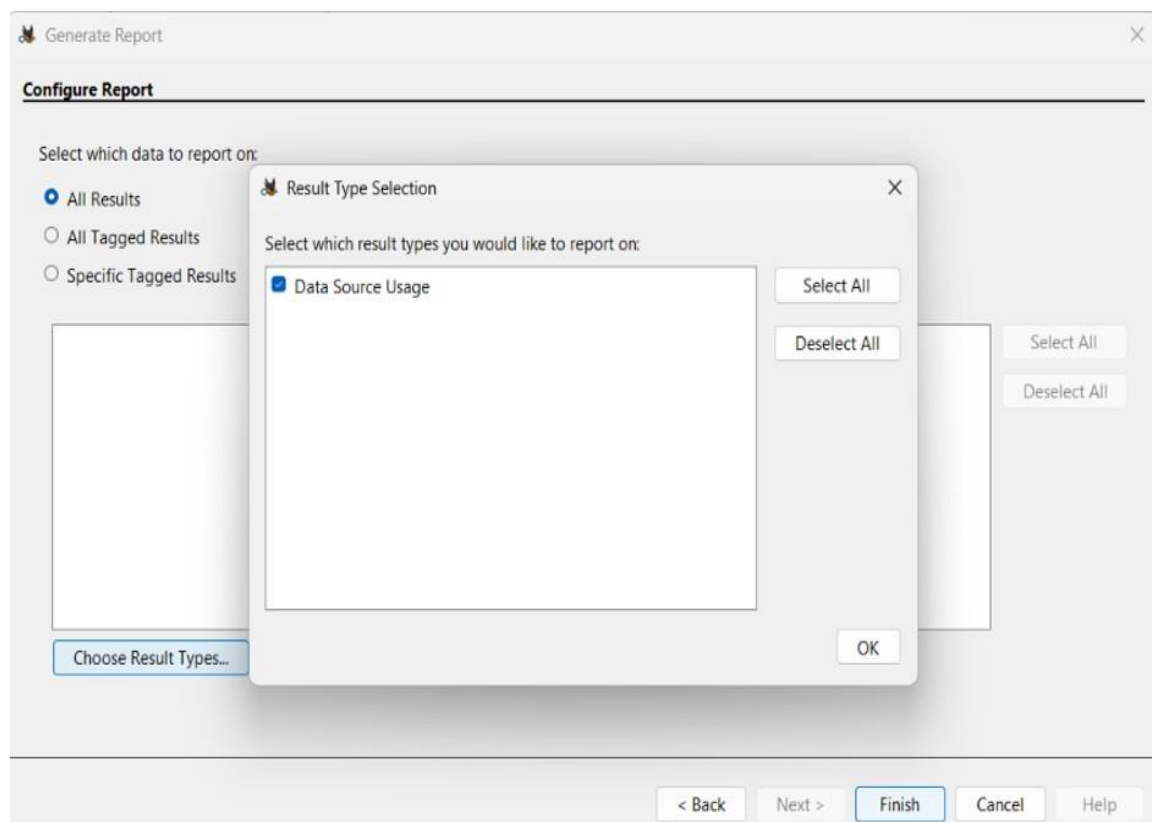


Figure 8: Configuring the report by selecting specific tagged data

RUBRICS:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

5. RESULT

The digital forensics experiment was successfully completed. A new case was established in Autopsy, and the provided .E01 disk image evidence was successfully imported, analyzed using ingest modules, and documented via the reporting feature.